

EE-312L Signals and Systems Lab
Laboratory Session 03
**Understanding The Basic Signals &
Classifications Using MATLAB**

October 9, 2024

Outlines:

1. Objective
2. Continuous-Time Signals and Discrete-Time Signals
3. Analog Signals and Digital Signals
4. Periodic Signals and Aperiodic Signals
5. Even Signals and ODD Signals
6. Symbolic Variables and Expressions in OCTAVE/MATLAB
7. Energy Signals and Power Signals

1 Objective

The objective of this lab session is to introduce basic signals and sequences using MATLAB.

2 Signals

- A signal is a flow of information.
- Mathematically, it is represented as a function of independent variables such as time (e.g. speech signal), position (e.g. image), etc.
- A common convention is to refer to the independent variable as time, although may in fact not.

Signal	Horizontal Axis	Vertical Axis
Analog	Continuous	Continuous
Discrete-Time	Discrete	Continuous
Digital	Discrete	Discrete

Table 1: Signal classification based on discrete and continuous values

2.1 Examples of Signals

- Speech: 1-Dimension signal as a function of time $s(t)$
- Grey-scale image: 2-Dimension signal as a function of space $i(x, y)$

2.2 Types of Signals

The independent variable may be either continuous or discrete

- Continuous-time signals
- Discrete-time signals are defined at discrete times and represented as sequences of numbers

The signal amplitude may be either continuous or discrete

- Analog signals: amplitude is continuous
- Digital signals: Amplitude is discrete

Computers and other digital devices are restricted to discrete time Signal processing systems classification follows the same lines.

3 Signal Processing

Modifying and analyzing information. Representation, transformation and manipulation of signals and information they contain.

3.1 Applications of Signal Processing

- Speech processing
- Enhancement – noise filtering
- Coding
- Text-to-speech (synthesis)
- Recognition
- Image processing

- Enhancement, coding, pattern recognition (e.g. OCR)
- Multimedia processing
- Media transmission, digital TV, video conferencing
- Communications
- Biomedical Engineering
- Navigation, radar, GPS
-

4 Continuous-Time Signals and Discrete-Time Signals

4.1 Continuous-Time Signals

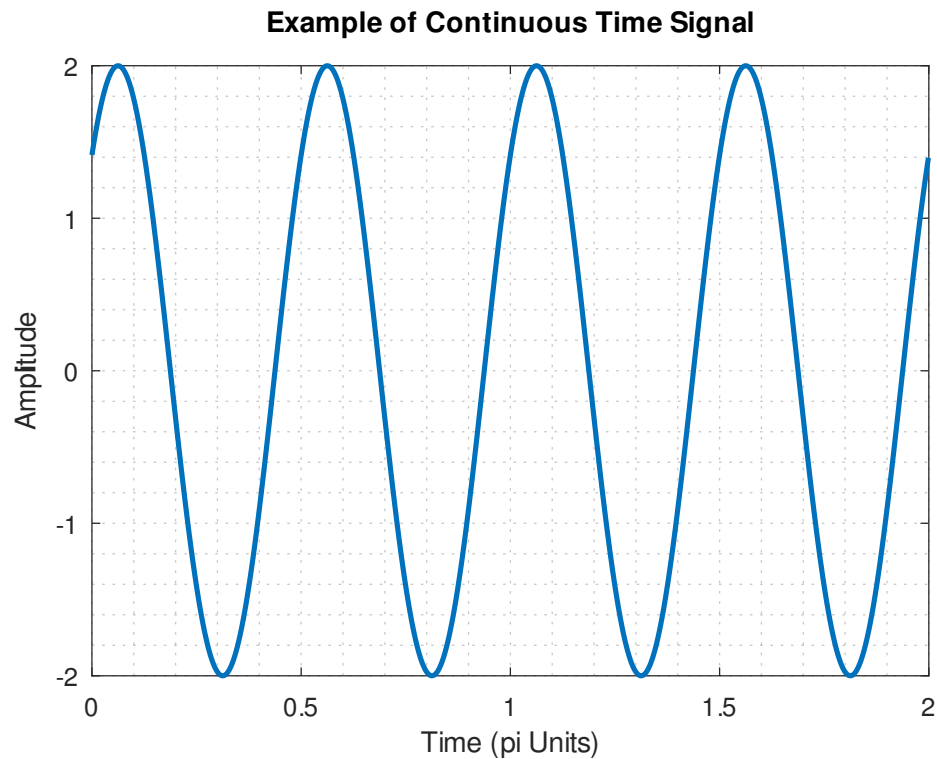
Continuous-Time Signal is the one whose domain is uncountable, we don't care about range. For storing the continuous time signal you need infinite memory which is not possible in any real system. So, finite no of samples of a continuous time signals are stored. For plotting the continuous time signals we use plot command, which simply connects the values of intermediate samples, gives us the illusion of continuous time signals.

```
1 A=2; % Amplitude
2 f=2; % Frequency
3 phase = pi/4; %Phase
4 t=(0:0.01:2*pi)/pi; %Horizontal time axis
5 y= A*sin(2*pi*f*t+ phase) %Vertical axis
6 plot(t,y, 'LineWidth',2.0);
7 grid("minor","on");
8 title('Example of Continuous Time Signal');
9 xlabel ('Time (pi Units)');
10 ylabel ('Amplitude');
```

4.2 Discrete Time Signals

Discrete Time Signals are those signals which are purely based upon domain, whose domain is Countable. We don't care about range of the signal.

```
1 A=1; % Amplitude
2 f=1; % Frequence
3 phase= pi; %Phase
4 n=0:10; %Sample Number
5 y= A*sin(f*n+ phase);
```



```
6 stem(n, y, 'LineWidth',3);  
7 title('Example of Discrete Time Signal');  
8 xlabel ('n (Sample Number)');  
9 ylabel ('Amplitude');
```

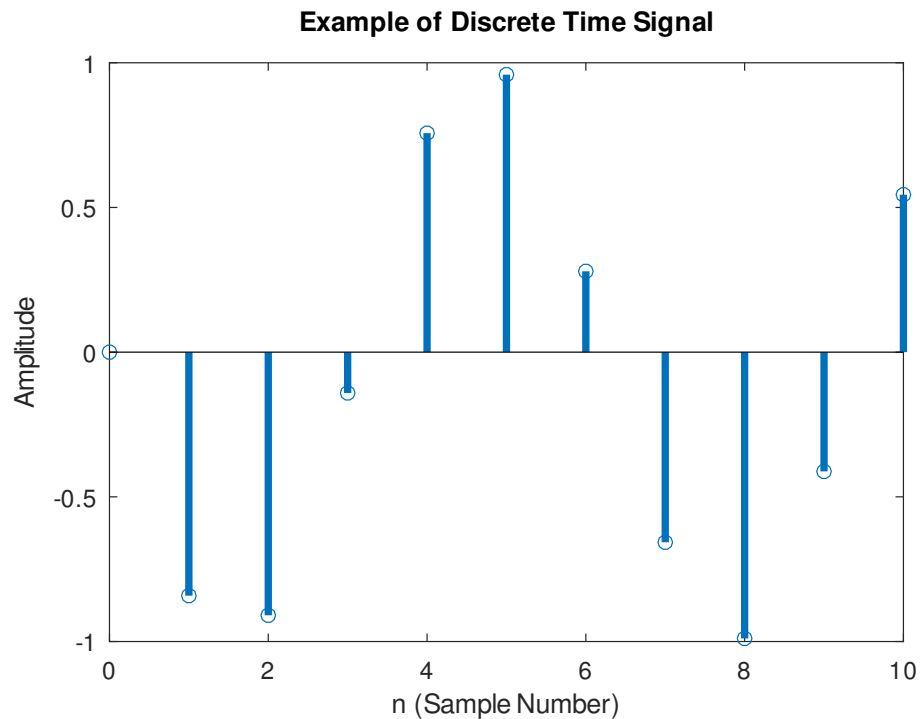
5 Analog Signals and Digital Signals

This classification of signal is purely based on range, we don't care about the domain.

5.1 Analog Signals

Analog Signals can take infinite values in range. In MATLAB we can plot any Analog signal for example $A \sin(2\pi ft + \phi)$ using plot command as follows:

```
1 % Analog Signal Example  
2 A=2; %Amplitude  
3 f=2; %Frequency  
4 phase=pi; %Phase  
5 t=(0:0.01:2*pi)/pi; %Time axis  
6 y= A*sin(2*pi*f*t+phase);  
7 plot(t, y, 'LineWidth',2);
```

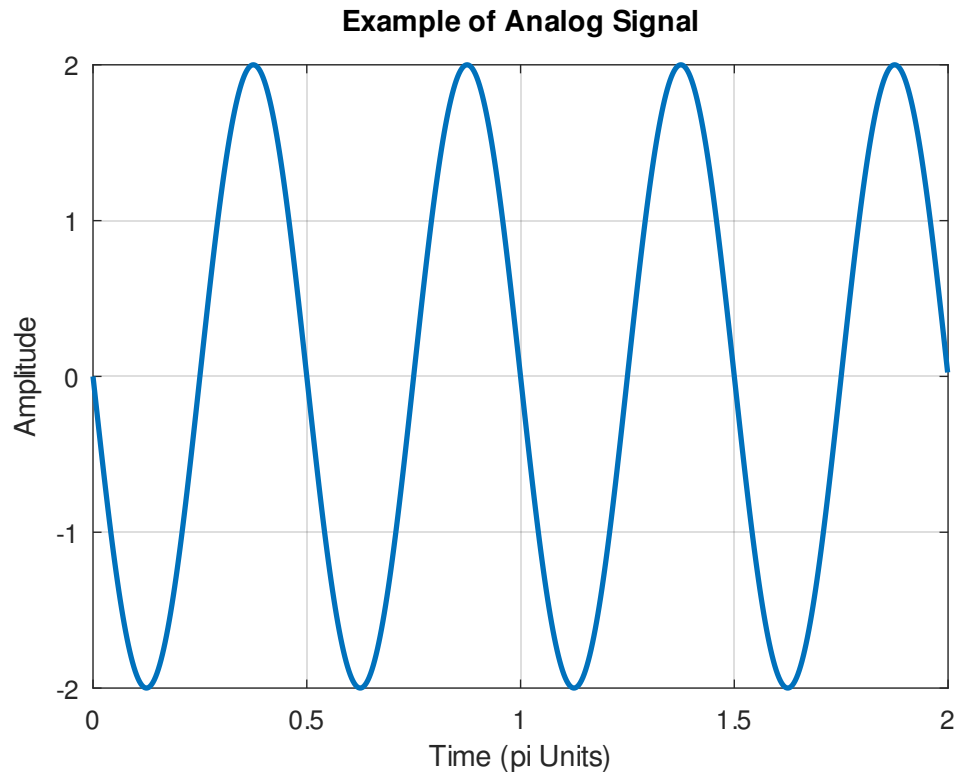


```
8 grid on;  
9 title('Example of Analog Signal');  
10 xlabel ('Time (pi Units)');  
11 ylabel ('Amplitude');
```

5.2 Sampled Version of Analog Signal (Discrete-Time Signal)

We can also plot analog signal using stem command.

```
1  
2 A=2; %Amplitude  
3 w=0.1; %Frequency  
4 phase = pi; %Phase  
5 n= (0:100); %Time axis  
6 y = A*sin(w*n+phase);  
7 stem(n, y, 'Linewidth', 2);  
8 title('Sampled Version of Analog Signal');  
9 xlabel ('Sample Number (Discrete)');  
10 ylabel ('Amplitude (Continuous)');
```



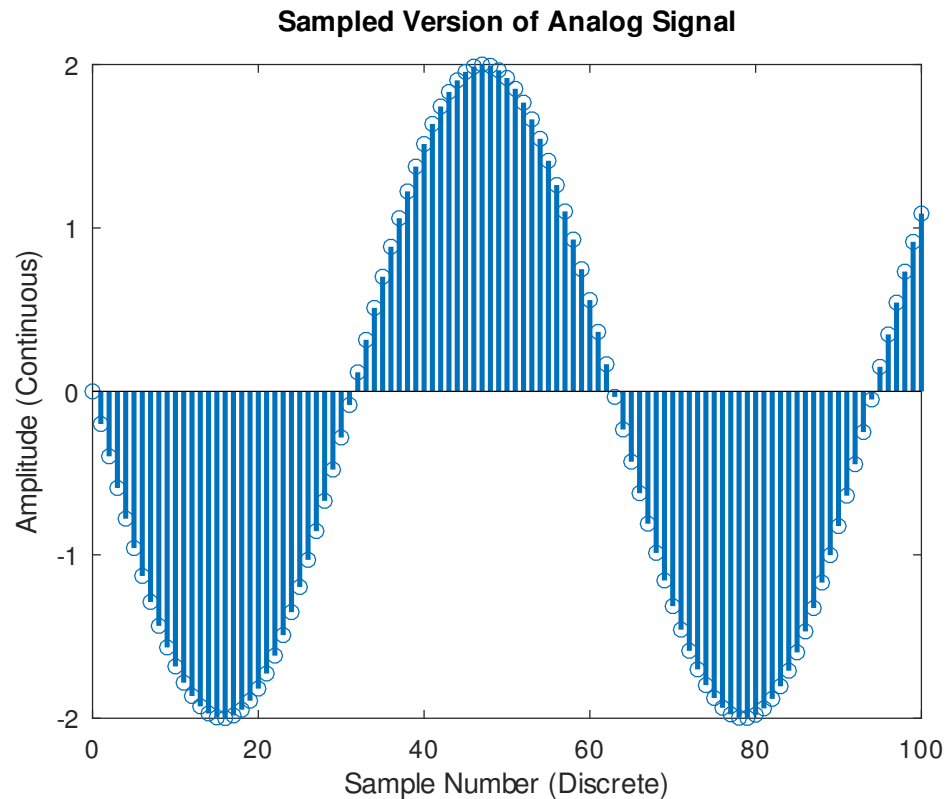
5.3 Digital Signal

Digital signal is the one which can take countable finite values in range. In MATLAB, normally, we can plot digital signal using stem or plot Commands.

```
1
2 A=2; %Amplitude
3 w=0.1; %Frequency
4 phase = pi; %Phase
5 n= (0:100); %Time axis
6 y = round(A*sin(w*n+phase));
7 stem(n, y, 'Linewidth', 2);
8 title('Example of Digital Signal');
9 xlabel ('Sample Number (Discrete)');
10 ylabel ('Amplitude (Discrete)');
```

Square wave is another example of digital signal.//

```
1 A=0.8; %Amplitude
2 f=2; %frequency
3 phase=pi; %Phase
4 t=(0:0.01:2*pi)/pi; %Horizontal time axis
5 y = A*square(2*pi*f*t+phase)
6 plot(t, y, 'Linewidth', 2);
```



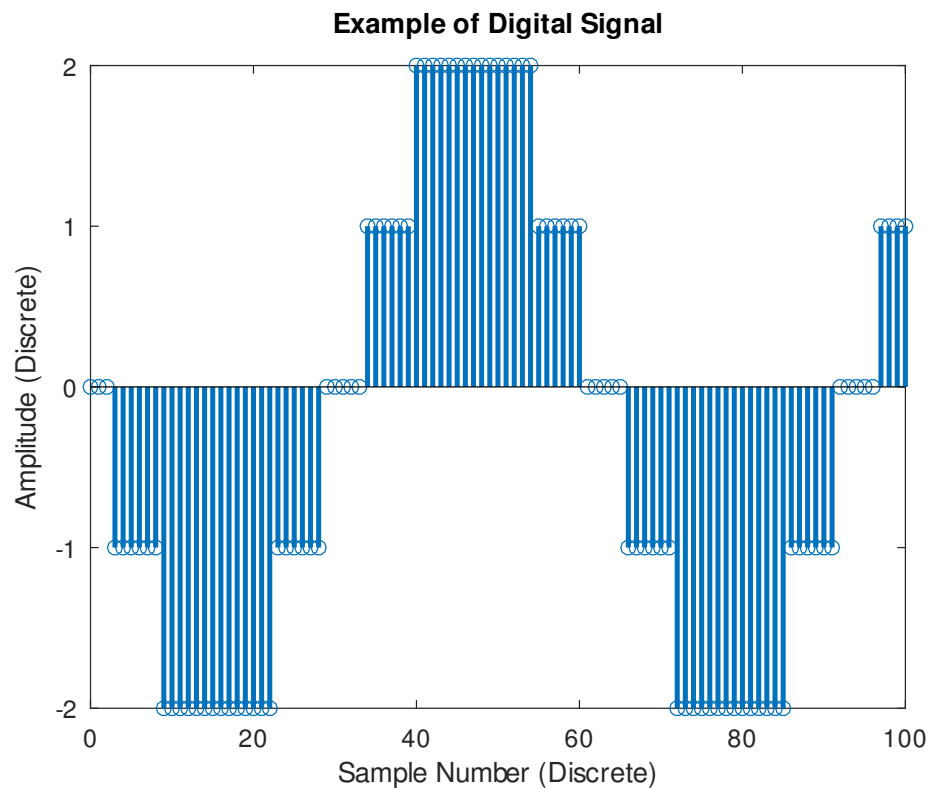
```
7 minor_grid on;  
8 axis([-0.1,2.1,-1.2,1.2]);  
9 title('Example of Digital Signal');  
10 xlabel ('Time (pi Units)');  
11 ylabel ('Amplitude');  
12  
13 ##T = 1/10;  
14 ##t = linspace(0,T,1001);  
15 ##y = square(2*pi*50*t);  
16 ##plot(t,y);  
17 ##axis([0 0.1 -1.5 1.5]);
```

6 Periodic Signals and Aperiodic Signals

6.1 Periodic Signals

A signal which repeats its pattern after a specific interval of time is called periodic signal.

Continuous-time periodic signal holds



$$x(t \pm T) = x(t), \quad (1)$$

where T is the fundamental time period.

```

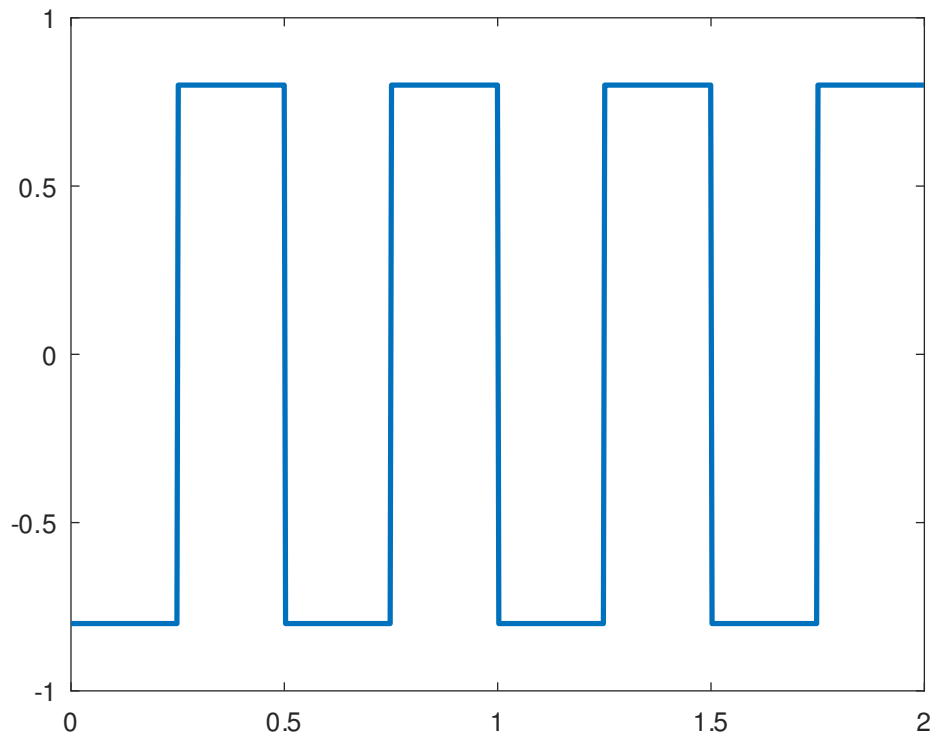
1 A=2; %Amplitude
2 f=1; %Frequency
3 t=(0:0.01:4*pi)/pi; %Time axis
4 y=A*sin(2*pi*f*t);
5 plot(t,y, 'LineWidth', 2);
6 title('Example of Periodic Signal');
7 xlabel ('Time (pi Units)');
8 ylabel ('Amplitude');
9 grid on;

```

Discrete-time periodic signal holds

$$x[n \pm N] = x[n], \quad (2)$$

where N is the fundamental time period.



A square signal

6.2 Aperiodic Signals

A signal which does not repeat its pattern after a specific interval of time is called aperiodic signal.

6.3 Symbolic Variables and Expressions

Need to install symbolic toolbox for using symbolic variables.

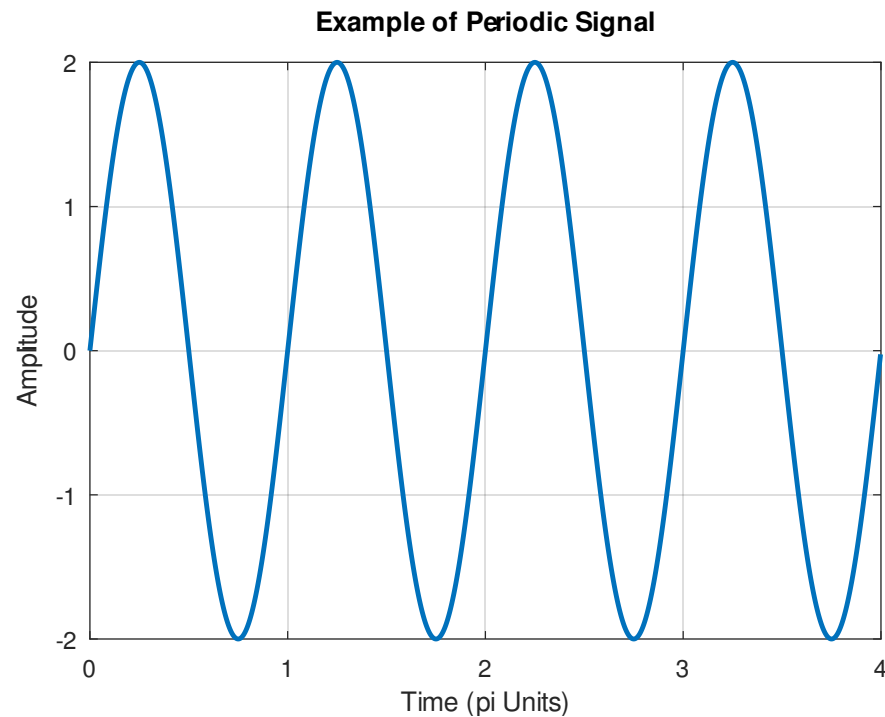
6.4 Symbolic Variables

Syntax:

```
syms arg1 arg2 ...
```

Example:

```
>>syms x; % create a symbolic object x  
>>syms x y z; % create several symbolic objects at once
```



Continuous-time periodic signal

6.5 Symbolic Expression

Example:

Declare the following

1. $f(x) = x \sin(x) + e^{-x/2}$
2. $g(x) = x \cos(x) - e^{-x/2}$

```
>>syms x; % define the symbolic variable x
>>f=x*sin(x)+exp(-x/2); % define the variable f
>>g=x*cos(x)-exp(-x/2); % define the variable g
```

6.6 Integration

`int()`

Syntax `int()`:

```
%*****indefinite integral: exp is expression****
int(exp)
%*****definite integral with limits a and b ***
int(exp, a,b)
%**** For more than one variables needs to mention ****
```

```
%**** the variable for integration ****
int(exp, var)
int(exp, var, a,b)
```

Example:

Evaluate the following using symbolic integration

1. $\int x \sin^2 x dx$

2. $\int_1^2 x \sin^2 x dx$

```
>> syms x;
>>int(x*sin(x)^2)
>>int(x*sin(x)^2, 1,2)
```

Example: integrate $x^3 + 3x^2 + 4x + 9$

```
%*****
syms x;
y=x^3+3*x^2+4*x+9;
I=int(y)

%*****Result*****
I =
x^4/4 + x^3 + 2*x^2 + 9*x
%*****
```

Example: integrate $x^3 + 3x^2 + 4x + 9$ over interval 3 to 5.

```
%*****
syms x;
y=x^3+3*x^2+4*x+9;
I=int(y,3,5)

%*****Result*****
I =
284
%*****
```

Example6: integrate $x^2 + 3xz + 1/2z^2 + z$ with respect to (w.r.t) x and z.

```
%*****
syms x;
y=x^2+3*x*z+1/2*z^2+z;
Ix=int(y,x)
Iz=int(y,z)
```

```
%*****Result*****
Ix =
x^3/3 + (3*z*x^2)/2 + (z^2/2 + z)*x

Iz =
z^2*((3*x)/2 + 1/2) + x^2*z + z^3/6
%*****
```

6.7 ezplot()

MATLAB has a built-in plot command called **ezplot()** that will plot symbolic functions with a single variable over some specified range.

```
>> ezplot(y, [ymin, ymax]) % plot the expression in y on the specified interval

>>syms x;
>>f=x*cos(x)+exp(-x/2);
>>ezplot(f, [-5,5]);grid
```

7 Energy Signals and Power Signals

An important classification of signals is between (finite) energy and (finite) power signals. The energy E_x of a continuous-time signal $x(t)$ is computed according to

$$E_x = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt \quad (3)$$

A signal $x(t)$ is an energy signal if its energy is definite, i.e., $0 < E_x < \infty$.

The power P_x of a signal $x(t)$ is given by

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt \quad (4)$$

If $x(t)$ is periodic, the power is computed from the relationship

$$P_x = \frac{1}{T} \int_{t_0}^{t_0+T} |x(t)|^2 dt \quad (5)$$

A signal $x(t)$ is called a power signal if $0 < P_x < \infty$.

The energy and power of a discrete-time signal are computed from quite similar expressions. The energy is given by

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2 \quad (6)$$

while the power is computed according to

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2 \quad (7)$$

7.1 Energy Signal

```

1 syms t T
2 x=2;
3 E=limit((int(x^2,t,-1,1)),T,inf)
4 P=limit((1/(2*T))*(int(x^2,t,-1,1)),T,inf)

```

Answer:

```

E = (sym) 8
P = (sym) 0

```

7.2 Power Signal

```

1 syms t T
2 x = t;
3 P = limit((1/(2))*int(x^2,t,-1,1)),T, inf)

```

Answer:

```

P = (sym) 1/3

```

Example: Power and Energy of continuous-time signal

Compute the energy and the power of the signal $x(t) = 2 \cos(\pi t)$ in one period time.

The period of $x(t) = 2 \cos(\pi t)$ is $T = \frac{2\pi}{\Omega} = \frac{2\pi}{\pi} = 2$

```

1 t1=0;
2 t2=2;
3
4 syms t
5 x = 2*cos(pi*t);
6
7 Ex = int(abs(x)^2,t,t1,t2)
8
9 Px = (1/(t2-t1))*int(abs(x)^2,t,t1,t2)

```

Answer:

Ex = (sym) 4

Px = (sym) 2

Example: Power and Energy of discrete-time signal

Compute the energy of the signal $x[n] = \cos(n)$, $-5 \leq n \leq 5$.

```
1 N=5;
2 n=-5:5
3 x=cos(n);
4 E = sum((abs(x)).^ 2)
5 P = 1/(2*N+1)*E
```

Answer:

E = 4.9058

P = 0.4460

8 Lab 03: Lab Tasks

1. Give two examples of Odd and Even signals each and plot them in MATLAB/OCTAVE.
2. Evaluate the following integrals using MATLAB/OCTAVE
 - $\int_0^1 x e^x dx$
 - $\int_0^1 x e^x \sin x dx$
3. The commands **diff()** and **solve()** are used for differentiation and solving algebraic expressions receptively. Give the syntax of **diff()** and **solve()** and illustrate each with an example. Also plot the examples using command **ezplot()**.
4. What is the difference between **ezplot()** and **ezsurf()** and differentiate by giving a plot example.
5. List 4 commands related to symbolic toolbox, that has not been discussed in this sheet.
6. Find the energy and power of the following signal

$$x(t) = \begin{cases} 2, & |t| \leq 1 \\ 0, & |t| > 1 \end{cases}$$